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Hardware

Nicla Vision

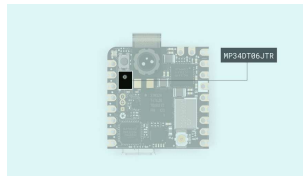
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Reading Audio Samples With the Onboard Microphone

Learn how to create a soundmeter using the built-in microphone with the Nicla Vision.

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Nicla Vision - microphone

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Overview

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Overview

In this tutorial you will use the **Arduino Nicla Vision** board to get the microphone (MP34DT06JTR) readings and change the LED brightness.

Goals

Get the microphone data

Use the PDM(Pulse-density modulation) library

Print the microphone values in the Serial Monitor

Change RGB blinking speed with the last microphone reading (Arduino IDE)

[Help](#)

openMV)

Required Hardware and Software

Nicla Vision

Latest mbed Core version

Latest openMV IDE version

Set Up

To check that you correctly set up the board, please visit our [Getting Started Guide](#) for both **OpenMV** and **Arduino** instructions.

Instructions

OpenMV

Open the script by going to **Examples > Arduino > NanoRP2040 > Audio > Audio_fft.py**.

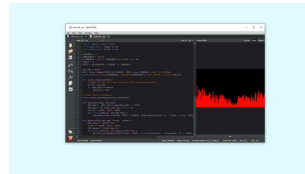


Using the same sketch as the NanoRP2040, because both boards access the microphone in the same way

Make sure the board is connected, if the board is connected to OpenMV you should see a green play button in the bottom left corner of the window. If you do not see this icon, try pressing the connect button in the bottom left corner. If there still is some issue to connect the board, take another look at the getting started guide.

When the script is running, you will

right panel that reflects the audio readings input. Try making some noise and see how it reacts.



OpenMV IDE - Spectrum analyzer

Arduino

Setting Up the Sketch

You will edit the example from the mbed Core, go to **Examples > PDM > PDMSerialPlotter** and save it into your sketchbook.

You can run the sketch to see the result, it will show the data that the microphone is getting on the **Serial Plotter**.

Controlling the Blinking LED

Now that you can get the microphone data, let's control the built-in RGB LED and change the speed of its blinking depending on the values, by changing the blinking time to the last reading of the microphone; the blink will be slow if the sound is loud, and fast if it is quiet.

You can access the example sketch at **Examples > PDM > PDMSerialPlotter** and then edit as shown in this tutorial. Or find the full edited sketch in our **Arduino_Pro_Tutorials** library.

```
1  /*
2     This example reads
3     out the samples to
4     Arduino IDE can be
5     Circuit:
6     - Arduino Nicla Vis
7     - Arduino Nano 33 E
8     - Arduino Nano RP20
9     - Arduino Portenta
10    This example code i
11    */
12
13
14    #include <PDM.h>
15
16    // default number of
17    static const char cha
18
19    // default PCM output
20    static const int freq
21
22    // Buffer to read sam
23    short sampleBuffer[51
24
25    // Number of audio s
26    volatile int samplesF
27
28    // Blinking
```

Testing It Out

After you have successfully verified and uploaded the sketch to the board, open the Serial Monitor from the menu on the left. You will now see the new values printed.

If you want to test it, the only thing you need to do is to speak or play some sounds close to the board and see how the blinking of the RGB LED changes based on the input.

Troubleshoot

In case the Serial Monitor

again. Now try to upload the sketch.

If the sketch is not working, try to double tap the reset button and upload the sketch once again.

Conclusion

You have learned how to use the Arduino IDE and OpenMV to get data from the microphone and then use it to change the RGB LED on the board. This can for example be used as an alarm system to wake the board up and take a screenshot with the Camera.

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